

Lists and indexing

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Loops and trace tables

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace an update for Lists and indexing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Lists and indexing. Proposed explanation: “Length is the last nonnegative index”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Distinguish a value from an index

Predict the three outputs. If the second print uses `values[3]` instead, where does execution stop and which later output is not produced?

```
values = [4, 9, 2]
print(values[1])
print(values[-1])
print(len(values))
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Separate index and value

For `counts = [4, 7, 4]`, give the length, valid nonnegative indices and value at index 1. Explain how the two items with value 4 differ.

My reasoning and evidence

6. W2 Trace an update

Start with counts = [4, 7, 4]. Run counts[0] = 6. Write the resulting list, its length and the value at index 2. Predict before running.

My reasoning and evidence

7. W3 Repair a position mistake

A learner loops with for count in counts and tries to read counts[count] for counts = [4, 7, 4]. Identify the first attempted index and error. Explain how to print each value directly.

My reasoning and evidence

8. W4 Process new data independently

Start with readings = [3, 0, 3, 2]. Replace the second item with 5 using its index. Write a loop that adds all updated readings and prints one final total. Trace every update of the total. Then test the summing part alone on [] and [6].

My reasoning and evidence

9. W5 Keep the information you need

Compare $[1, 5, 2]$ and $[5, 1, 2]$. State their totals and first items. Would storing only the total answer which first item was larger? Explain.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled lists and indexing evidence sheet to Python readiness portfolio. Show how you can distinguish a list index from the value stored at that position. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Lists and indexing

Indexes are positions, not values. A list of shelf counts is read by position.

Python 3: `counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).`

Zero-based list positions

	Index 0	Index 1	Index 2
Stored counts	6	1	8

Read a cell using both its row and column labels.

Lists and indexing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw positions

Indexes 0,1,2 contain 6,1,8 respectively.



2. Read two positions

Index 0 returns 6; index -1 returns the final value 8.



3. Check length boundary

Length is 3, but index 3 is beyond the final valid positive index 2.

CHANGE ONE THING

Replace the second print with `print(counts[3])`.

Explain the changed result using the matching lesson.

Label the three index positions. Predict all original outputs, then name the first output and stopping point after using index 3.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Indexes are positions, not values. A list of shelf counts is read by position.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Length is 3, but index 3 is beyond the final valid positive index 2.

1. Read two positions
2. Check length boundary
3. Draw positions

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check length boundary Length is 3, but index 3 is beyond the final valid positive index 2. List positions begin at zero. A negative index counts from the end; the list length counts entries and is not itself a valid positive index for the final entry.

Practice 2: explain the deciding evidence

Changed-case practice: Replace the second print with `print(counts[3])`. Which conclusion is supported by the supplied facts?

1. The first print shows 6, then `IndexError` stops the script before the length print.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The first print shows 6, then `IndexError` stops the script before the length print. The attempted position does not exist.

Repair a missing building block

Review Loops and trace tables (python-bridge/6). Goal: Trace variable values before and after each loop iteration. Short practice: List the four visited values and their running totals. Explain why 5 is absent and what prints for `range(1,1)`. Return to this week and explain what you changed.

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16. Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual

results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
